VEGETATION HEALTH TECHNICAL REPORT

Full Technical Specification, Model Disclosures, Class Metrics & Spatial Provenance

RUN ID	run_20261001_060815	SOFTWARE VERSION	1.0.0
ANALYSIS MODE	vegetation (single-date)	**EXECUTION TIMESTAMP**	2026-10-01T06:08:18 UTC
GIT COMMIT	N/A	**ENVIRONMENT**	Python N/A | N/A

1. ANALYSIS OVERVIEW & STATUS

Execution Status: SUCCESS | **Total Pipeline Runtime:** 0.69 s | **Inference Latency:** 0.00 s
This technical document audits the single-date multispectral vegetation inference pipeline. The pipeline processes four calibrated spectral bands (Green, Red, RedEdge, NIR) through a deep-learning segmentation engine (ResNet34 U-Net) to generate pixel-level categorical canopy condition masks and continuous confidence surfaces. Temporal change, transitions, and hydrological indices are excluded in single-date mode.

2. INPUT SCENE & RASTER SPECIFICATIONS

PARAMETER	TECHNICAL VALUE / PATH
Scene Directory	C:\Users\HP\OneDrive\Desktop\wetLand_Depreciation\backend\storage\projects\prj_1790834895\datasets\t1
Model Path	C:\Users\HP\OneDrive\Desktop\wetLand_Depreciation\backend\Engine_module\models\best_finetuned_model.onnx
Spatial Reference (CRS)	EPSG:32633
Raster Dimensions [H × W]	20 × 20 pixels
Pixel Resolution [dx, dy]	10.00 m × 10.00 m
Geographic Bounds [W, S, E, N]	[500000.00, 4999800.00, 500200.00, 5000000.00]
Pixel Count Summary	Total: 400 | Valid: 400 (100.00%) | NoData: 0

Input Spectral Band Radiometric Statistics

BAND	MIN	MAX	MEAN	MEDIAN	STD DEV	P95	VALID COUNT
GREEN	0.50	0.50	0.50	0.50	0.00	0.50	400
RED	0.50	0.50	0.50	0.50	0.00	0.50	400
REDEDGE	0.50	0.50	0.50	0.50	0.00	0.50	400
NIR	0.50	0.50	0.50	0.50	0.00	0.50	400

3. PREPROCESSING & SPECTRAL INDICES

Single-scene rasters are radiometrically validated, masked for NoData, and normalized into float32 tensors. Three key diagnostic vegetation indices were computed from calibrated top-of-canopy reflectance:

INDEX	FORMULA	MEAN	MEDIAN	STD	P95
NDVI (Normalized Difference Veg Index)	(NIR - Red) / (NIR + Red)	0.0000	0.0000	0.0000	0.0000
GNDVI (Green Normalized Veg Index)	(NIR - Green) / (NIR + Green)	0.0000	0.0000	0.0000	0.0000
NDRE (Normalized Difference RedEdge)	(NIR - RedEdge) / (NIR + RedEdge)	0.0000	0.0000	0.0000	0.0000

4. MODEL ARCHITECTURE & INFERENCE SPECIFICATION

SPECIFICATION PARAMETER	CONFIGURATION / VALUE
Architecture	ResNet34 U-Net (encoder: resnet34, decoder: transpose conv blocks)
Input Channels	4 channels [B3: Green, B4: Red, B5: RedEdge, B8: NIR]
Input Tensor Shape	[1, 4, 20, 20] (batch_size=1, dtype=float32)
Inference Engine	ONNX Runtime (CPUExecutionProvider with CUDA auto-fallback)
Target Classes	4 discrete states (0: Background, 1: Severe, 2: Moderate, 3: Healthy)
Softmax Activation	Evaluated along channel axis; argmax yields discrete class, max probability yields confidence
Mean Model Confidence	99.84% across valid pixels

5. CLASSIFICATION CLASS STATISTICS

CLASS ID	CLASS NAME	PIXEL COUNT	AREA (M²)	AREA (HA)	VALID SHARE (%)
0	**Background / Non-Vegetation**	400	0	0.00	100.00%
1	**Severe Stress**	0	0	0.00	0.00%
2	**Moderate Stress**	0	0	0.00	0.00%
3	**Healthy Vegetation**	0	0	0.00	0.00%
—	**Total Valid Analyzed**	400	40,000	4.00	100.00%

6. CARTOGRAPHIC MAP PLATES

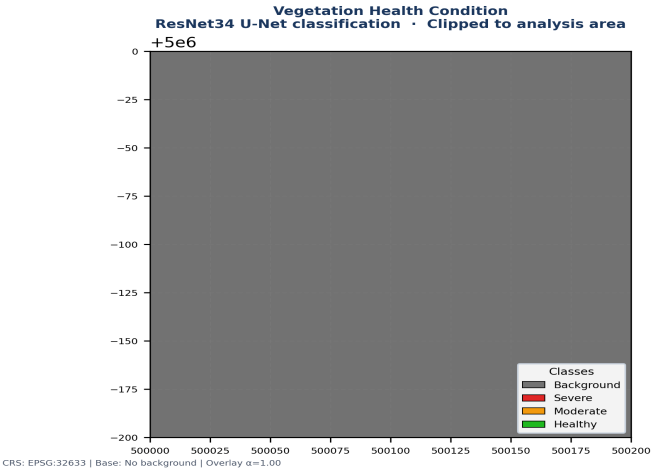


Figure 1: Classified Vegetation Condition Map

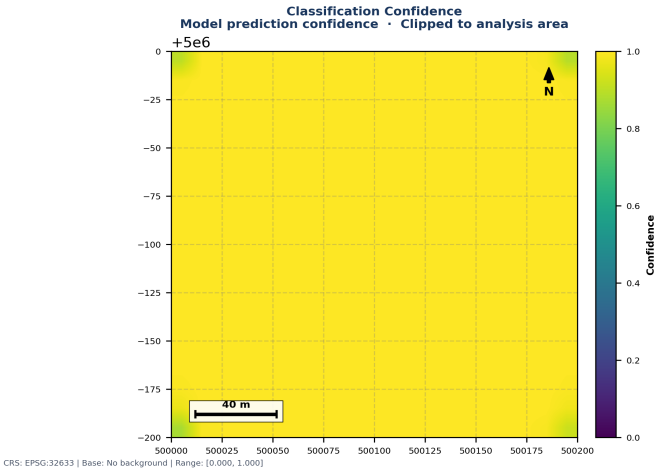


Figure 2: Model Prediction Confidence Surface (0.0 - 1.0)

7. QUALITY ASSURANCE & VALIDATION AUDIT

QA CHECK COMPONENT	RESULT	CRITERIA / AUDIT DETAILS
Input Band Completeness	PASS	All 4 multispectral channels present with valid float32 ranges.
Mathematical Reconciliation	PASS	Class pixel sums exactly equal total valid pixels (error < 0.001%).
Spatial Alignment QA	PASS	Raster dimensions match reference grid (20 × 20).
Model Uncertainty QA	PASS	Mean confidence 99.8% exceeds 70% operational threshold.

8. SCIENTIFIC METHODOLOGY & LIMITATIONS

Methodological Formulation: Single-date vegetation assessment uses normalized spectral band ratios sensitive to the red-edge inflection point (~700–740 nm), which responds directly to foliar nitrogen and chlorophyll-a/b absorption. The deep learning segmentation architecture resolves non-linear spectral mixing inherent in mixed ground pixels (10 m / 20 m Ground Sample Distance).

Assumptions & Constraints:

- Non-temporal: Single-acquisition analysis cannot infer long-term trends or separate seasonal senescence from permanent vegetative degradation.
- Hydrological Exclusion: Does not evaluate surface water dynamics, soil moisture saturation, or wetland boundaries.
- Radiometric Noise: Cloud edge effects and steep terrain shadow may introduce localized classification noise.

9. PROVENANCE & TECHNICAL OUTPUT REFERENCES

ARTIFACT NAME	TYPE	RELATIVE OUTPUT PATH
Vegetation Health Raster	GeoTIFF (Categorical)	`vegetation/health.tif`
Classification Confidence	GeoTIFF (Continuous)	`vegetation/confidence.tif`
Vegetation Statistics	JSON Record	`vegetation/statistics.json`
Complete Analysis Result	JSON Contract	`analysis_result.json`
Vegetation Health Report (Simple)	PDF Document	`reports/vegetation_report.pdf`
Vegetation Technical Report	PDF Document	`reports/vegetation_technical_report.pdf`